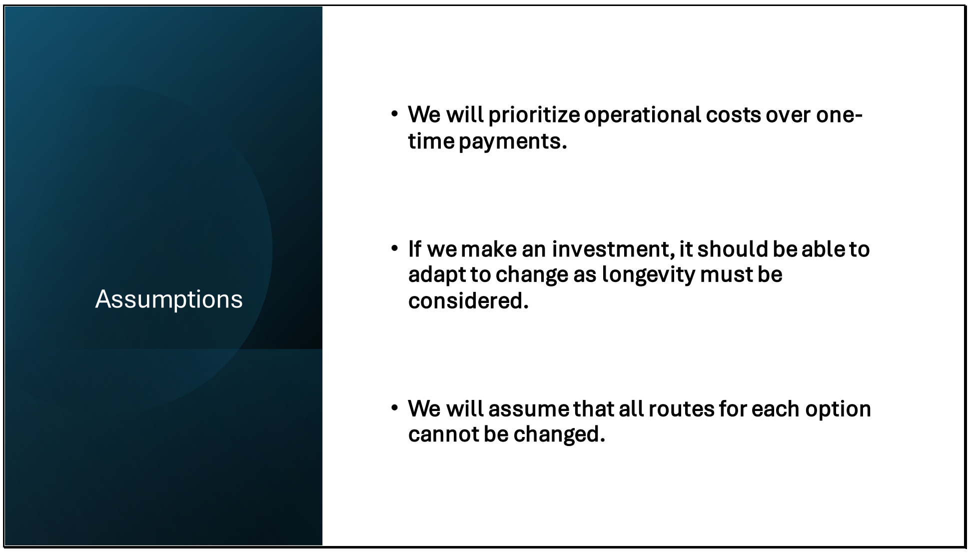
Slide 1

Our Take on the Grantshire Gritting Depot Expansion/Relocation Problem

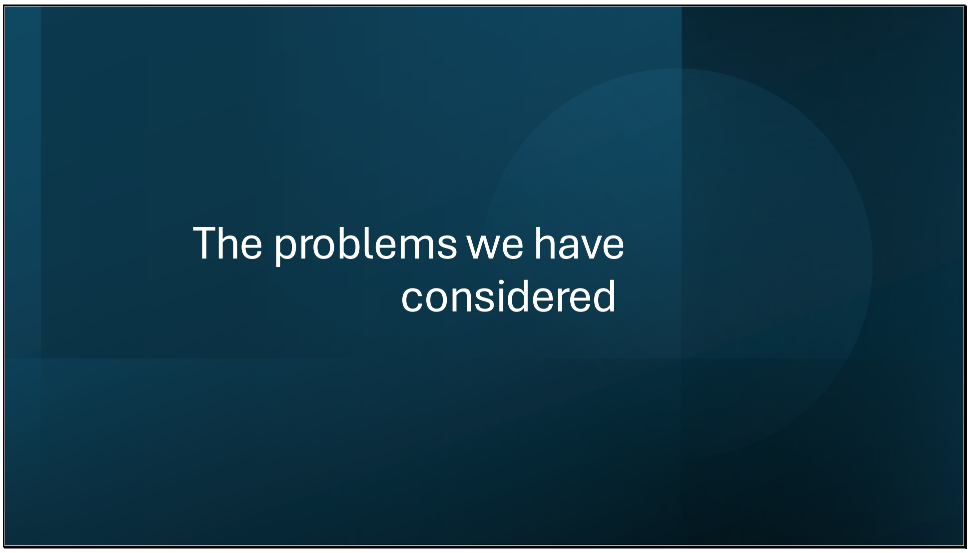
Presented By - ProConsult Solutions

Slide 2



There are three main assumptions which we have concentrated on whilst looking into each possible investment option. The first is the operational costs. Operational costs are vital to understand how expensive each option is as they help to understand the costs which will reoccur each year. This can help to choose the option which can suit the council in both the short and long term. The second assumption which we focused on is the fact the option must be able to adapt in order to ensure its longevity. This is important as without this the investment may be useless in the long term as the needs of the council change (option of buying new gritters, catered storage capability for grit salt, expecting the expansion of the number of houses and therefore roads(north because motorway south)). The final assumption is that the different routes taken cannot be changed as they have been calculated by the algorithm and are unlikely to be changed.

Slide 3



We will now talk about the various issues we have considered that hold weight in our final decision.

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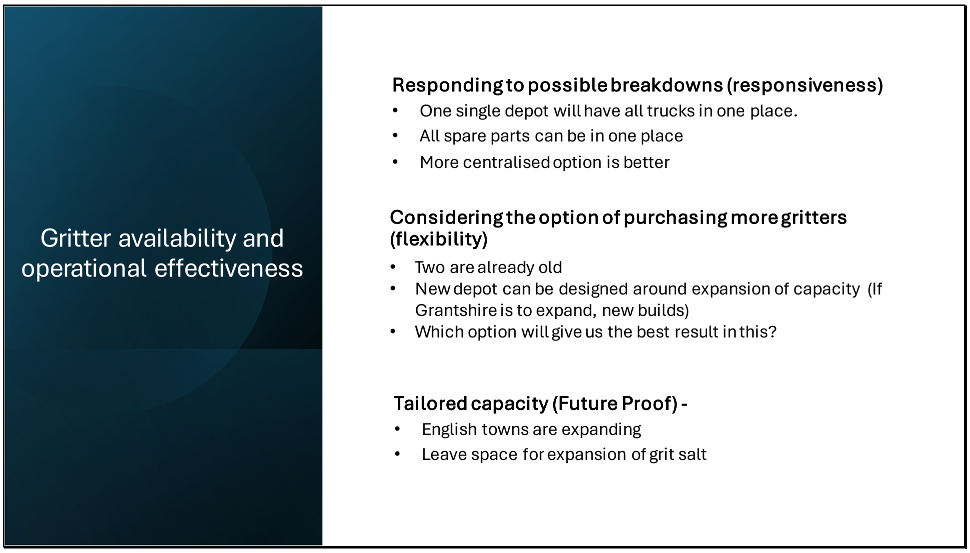

This investment into Grantshire gives us the opportunity to consider how we can best maintain and even optimise your fleet of gritting trucks. We must have a forward-thinking approach in all our choices as this investment is bound to develop over the long-term.

- Firstly, We must consider the scenario of possible breakdowns and which investment positions us to best respond to this. An option that gives us a more centralised operation is best for these scenarios as all spare parts and vehicles can be kept in one place.
- We must consider the possibility that more trucks could be needed in the future (as you currently have two coming to the end of their life cycle), we considered how we can house these extra trucks now to maximise this investment and mitigate the chance of future costs.
- Adding to this we must consider future proofing this investment by tailoring the storage to our current needs with space to expand in the event of the expansion of Grantshire, as many British towns are set to expand in the next few decades. In Grantshire’s case the town is likely to expand north meaning a focus should be put on the more northern depot options.

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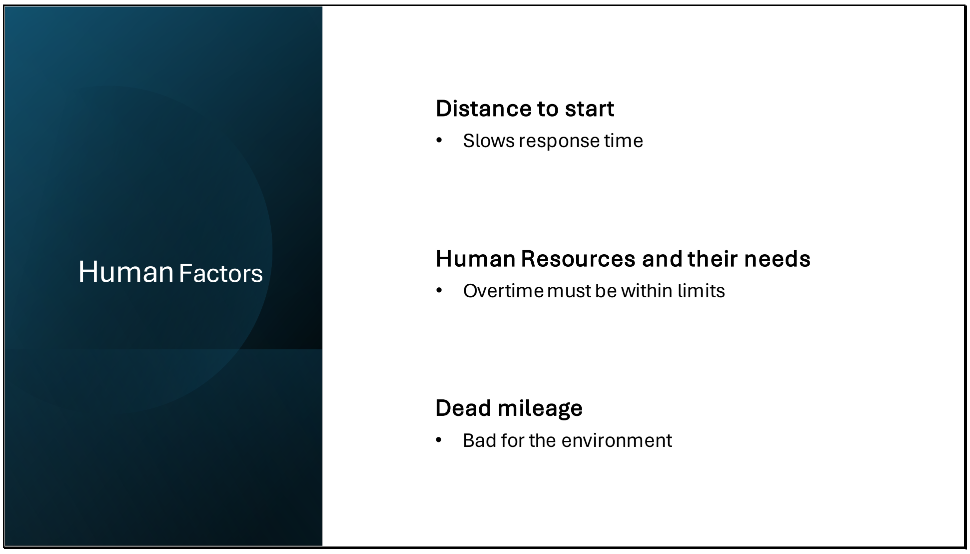
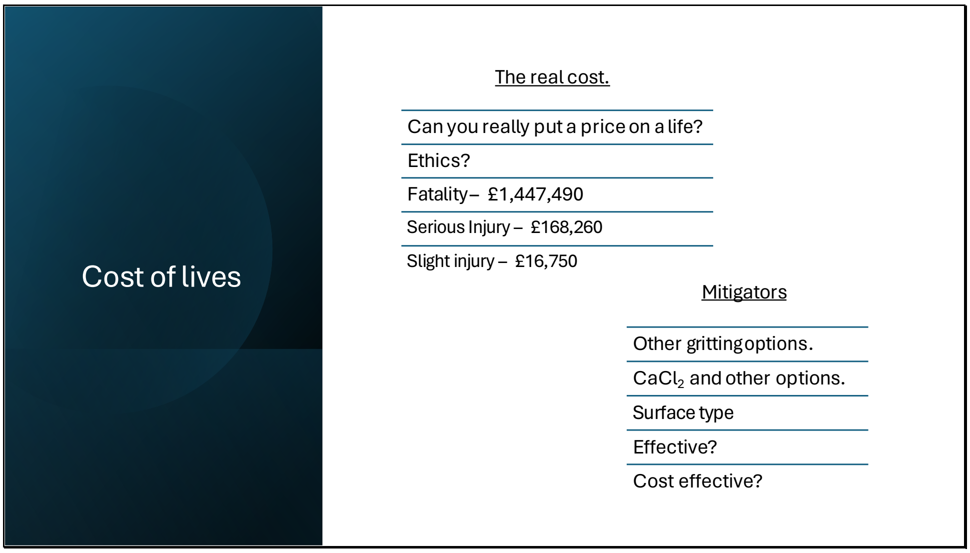


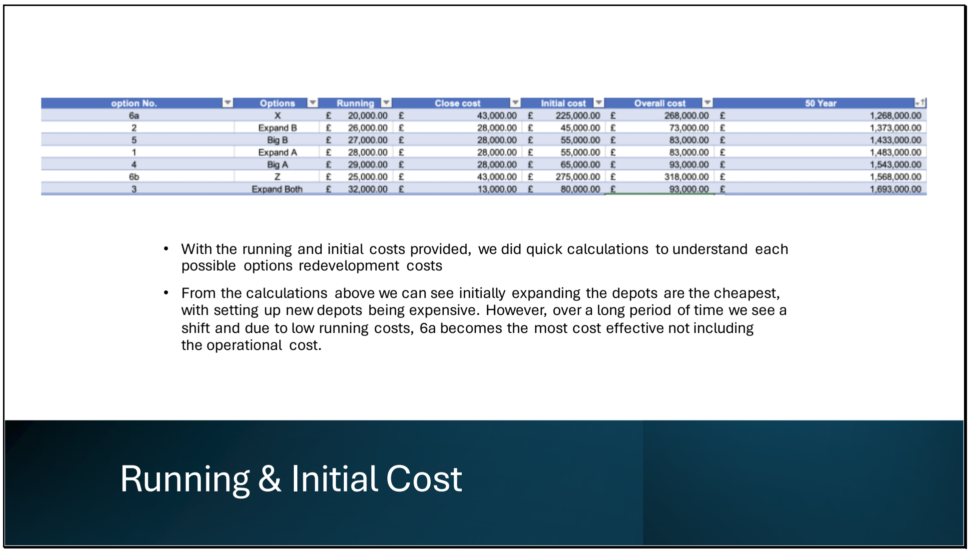
Another problem which we are facing is the Human Factors which the council must consider. The amount of time each member of staff works must be considered to ensure that they aren't overworked. The impact which the dead mileage has on the environment must be considered as increased wasted mileage will lead to a bigger impact on the environment as well as higher wasted fuel costs. Distance to start must also be considered as if the depot is centrally located then it is more likely to be closer to where the workers live as well as reducing the response time to adverse weather conditions.

Slide 6



Can you really put a price on a life? What's one worth to you? And do you value money over life saving? These are some of the questions to ask yourselves when thinking about the cost of lives.
This all ties into the ethics. The main question being: Is it right or wrong to cut costs at the risk of safety?
As you can see the cost of the outcomes in the case of an accident.
Most grit salt mixes consist of 90% NaCl yet there are alternatives.
One is CaCl2 which is more effective but considerably more expensive.
$1 vs $0.1 per gallon for Calcium Chloride and Sodium Chloride respectively.
Calcium Chloride breaks up into 3 ions instead of the 2 that NaCl does, this means that CaCl2 is considerably more effective at lowering the freezing point for water.
As for Cost effective that would depend on how much value you put on life.
With road surfaces changing between urban roads and rural roads new and old you need to consider if the type of salt could provide a difference in overall safety.

Slide 7



option No.	Options	Running	Close cost	Initial cost	Overall cost	50 Year
6a	X	£ 20,000.00	£ 43,000.00	£ 225,000.00	£ 268,000.00	£ 1,268,000.00
2	Expand B	£ 26,000.00	£ 28,000.00	£ 45,000.00	£ 73,000.00	£ 1,373,000.00
5	Big B	£ 27,000.00	£ 28,000.00	£ 55,000.00	£ 83,000.00	£ 1,433,000.00
1	Expand A	£ 28,000.00	£ 28,000.00	£ 55,000.00	£ 83,000.00	£ 1,483,000.00
4	Big A	£ 29,000.00	£ 28,000.00	£ 65,000.00	£ 93,000.00	£ 1,543,000.00
6b	Z	£ 25,000.00	£ 43,000.00	£ 275,000.00	£ 318,000.00	£ 1,568,000.00
3	Expand Both	£ 32,000.00	£ 13,000.00	£ 80,000.00	£ 93,000.00	£ 1,693,000.00

After looking at human related factors towards deciding which option is best, we looked at the running costs and initial costs given to us by Peter. We calculated the first overall cost with these figures which showed us that option 2 with the cheapest with options 5 and 1 being second cheapest being £10,000 more expensive. Therefore initially without the operational costs, option 2 would be the most cost effective. However after doing calculations for cost over a longer period of time we can see that due to lower running costs, 6a becomes the most cost efficient option beating the option 2 by £105,000. Over a long period of time such as the 50 years we calculated we determine that the difference in cost is not significant so based of these calculations we suggest that either option 6a or 2 is a suitable option.

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costing model based on 42 events

Options	1	2	3	4	5	6a	6b
Operational cost from costing model	£ 352,679.00	£ 351,519.00	£ 357,099.00	£ 353,915.00	£ 353,695.00	£ 346,359.00	£ 347,999.00
drivers extra cost	£ 18,480.00	£ 26,880.00	£ 21,840.00	£ 25,200.00	£ 31,920.00	£ 26,880.00	£ 16,800.00
Overall operational cost	£ 371,159.00	£ 378,399.00	£ 378,939.00	£ 379,115.00	£ 385,615.00	£ 373,239.00	£ 364,799.00

- For an average winter we chose a figure of 42 events, we chose this figure by doing forecasting past data. We did multiple moving averages and they all predicted number of events to be between 41-43.
- From the calculations we found that option 6b had the lowest operational cost with option 1 coming in second with a difference of over £6000 per year.
- As we can see from the table even though option 6b didn't have the lowest initial operational cost, due to them having a low driver additional cost, they became the cheapest.

Average Winter Findings

After looking at the initial costs, we wanted to then look at costs for an average winter. The number of events we chose for an average winter was 42. You may wonder how we got this figure? We did this through forecasting. After calculating multiple moving averages, all of them predicted events to be between 41 and 43. As from the calculations on the slide, we initially calculated for operational costs which include the new running costs and then we calculated the overtime costs for the drivers. From the figure on the slide the best option for operational costs was 6a with 6b following right behind it; this is due to these depots being completely new allowing us to have the newest technology to bring operational cost down. The lowest of the drivers cost was 6b and 1, due to this 6b become overall the cheapest option for overall operational cost, with option 2 being close second cheapest.

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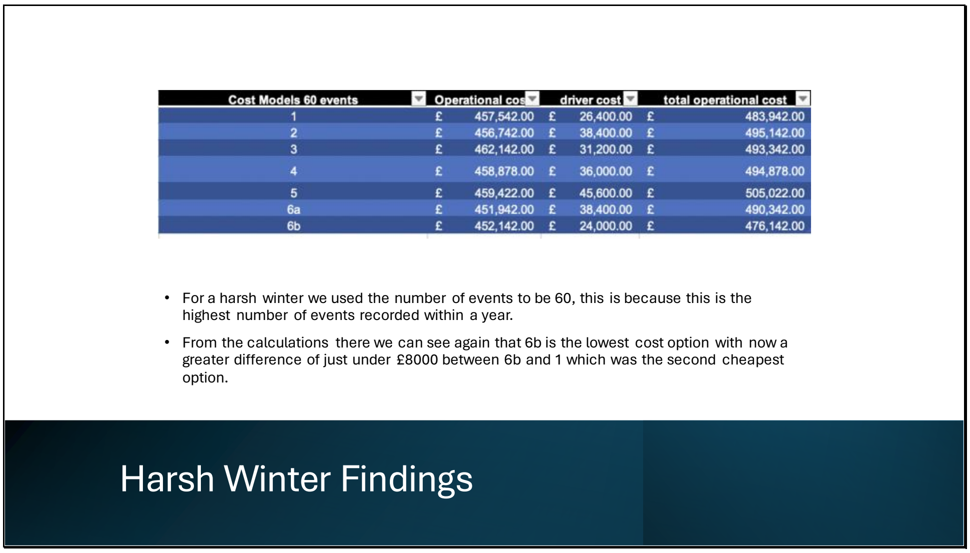

Cost Models 60 events	Operational cos		driver cost		total operational cost
1	£	457,542.00	£ 26,400.00	£	483,942.00
2	£	456,742.00	£ 38,400.00	£	495,142.00
3	£	462,142.00	£ 31,200.00	£	493,342.00
4	£	458,878.00	£ 36,000.00	£	494,878.00
5	£	459,422.00	£ 45,600.00	£	505,022.00
6a	£	451,942.00	£ 38,400.00	£	490,342.00
6b	£	452,142.00	£ 24,000.00	£	476,142.00

- For a harsh winter we used the number of events to be 60, this is because this is the highest number of events recorded within a year.
- From the calculations there we can see again that 6b is the lowest cost option with now a greater difference of just under £8000 between 6b and 1 which was the second cheapest option.

Harsh Winter Findings

Next we did calculations for a harsh winter, this is so we can present to you an example of worst case scenario and how much it would cost. You can see on the slide we have provided the results again. We can see that 6b is the lowest again due to the low operational and driver costs. Option 6b is £8000 cheaper than the next best option which was option 1, therefore we see both options to be very cost effective. However over a long period of time we will see the difference in costing increase. For example over 50 years the difference between these options will be £400,000 and we see this as a significant change and see 6b as the best cost effective option.

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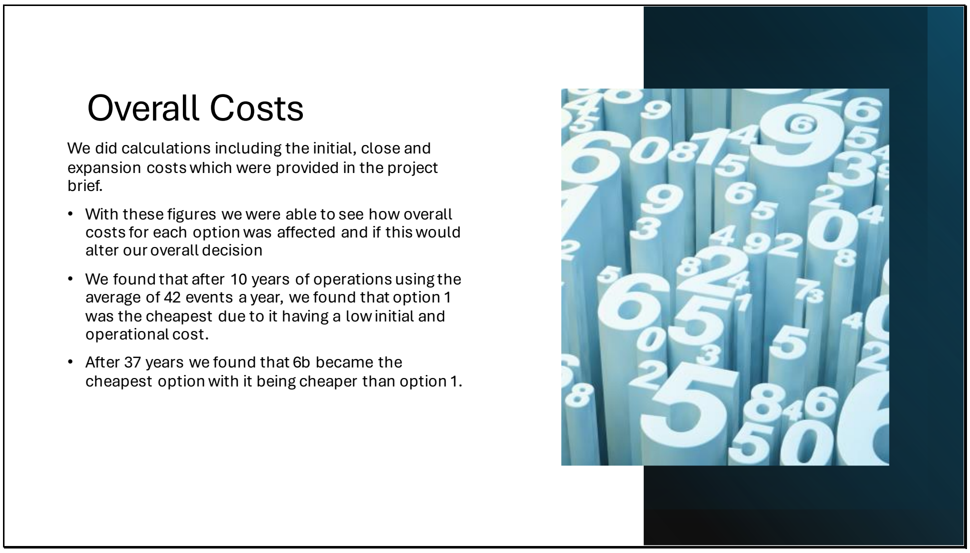

Overall, due to our calculations we could see the differences in the costs for options depending whether we were looking at initial cost, operational cost or cost over a longer period of time. From costs alone we came to the conclusion that the best suitable options are 6b and 1. Due to them having the lowest operational costs. We also determined that operational cost is a lot more suitable to determine the best option that initial cost due to them going away after a year. However when looking at the overall costs when looking at an average winter we found after 10 years option 1 was the cheapest due to low initial and operational cost. After 37 years we found 6b became the most cost effective option with and with this depot having the lowest operational cost, this option will always be the cheapest from this year onwards. Overall we determined from costs that 6b was the better option due to it having the lowest overall operational cost.

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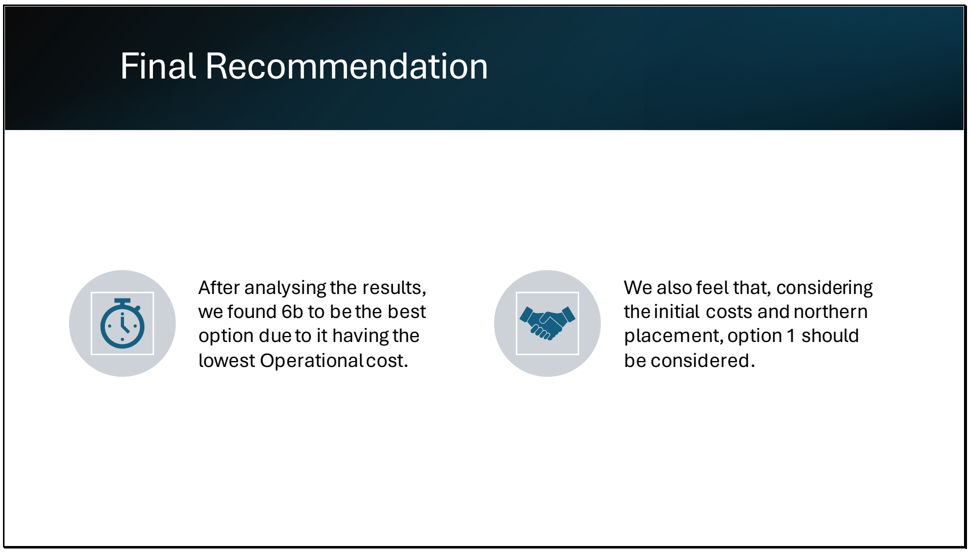

Having considered the problems and after analysing the results we found 6b to be the best option due to it having the lowest total cost across 50 years, it also had the lowest operational cost which demonstrates that this depot will be more efficient to use for the Grantshire Gritting.
We also feel that, considering the initial costs, option 1 should be considered. As Grantshire will likely expand north option 1 is effective as it focuses on depots a and b which are both situated near to the north making them ideal to react to this expansion.
For this model we have not considered alternative grit salts and would not recommend them as the cost is too much to be worth and will heavily effect operational costs.
We have managed to narrow it down to these 2 options and would recommend you invest in option 6B, but we would like to hear your thoughts and expertise on this matter.

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Thank you for listening

We are open to any questions